

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.1: The Periodic Table — Teacher Reference (Pre-filled)

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SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand	Sub-Strand 2.1: The Periodic Table
Driving Question	DRIVING QUESTION: Why do the fertiliser bags on Kenyan farms carry such specific and different chemical formulae — and what does the inner structure of an atom have to do with it?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Atomic Structure: Protons, Neutrons, and Electrons	Students examine labelled diagrams of atoms (hydrogen, carbon, nitrogen, phosphorus) and observe that each atom has a nucleus containing protons and neutrons, surrounded by electrons arranged in shells. Teacher notes: Use the WAIT TIME (10 seconds) after asking 'What do you notice is different between these atoms?' before cold-calling 3 students. Anchor to phenomenon: display a CAN 17:17:17 fertiliser bag and ask students what the numbers might mean.	Students learned that every atom is made of three subatomic particles — protons (positive charge, in nucleus), neutrons (no charge, in nucleus), and electrons (negative charge, in shells). The atomic number equals the number of protons, which also equals the number of electrons in a neutral atom. The mass number equals protons + neutrons. Pedagogical note: Reinforce that nitrogen (N, atomic number 7) and phosphorus (P, atomic number 15) — both found in fertilisers — serve as concrete	Students explain that the identity of an element is determined by its number of protons (atomic number). They connect this to the fertiliser phenomenon by reasoning that nitrogen atoms (7 protons) and phosphorus atoms (15 protons) are fundamentally different because of their inner structure — laying the groundwork for why different fertiliser formulae carry different elements. Teacher notes: Prompt with 'How does knowing the atomic number of nitrogen help us understand why CAN

		examples throughout this lesson.	and DAP are different fertilisers?' Cold-call 2 students to summarise.
Lesson 2 Electron Configuration and the Periodic Law: Finding the Hidden Pattern	Students observe the electron shell configurations of the first 20 elements (H to Ca) arranged on a display card, noting how the number of outer-shell (valence) electrons repeats in a regular pattern across periods. Teacher notes: Direct students to count valence electrons for Li, Na, K (all 1) and Be, Mg, Ca (all 2). WAIT TIME of 15 seconds after 'What repeats as you move across each row?' before cold-calling 3 students. Connect to fertiliser bag: nitrogen is in Period 2, phosphorus in Period 3 — ask students what that tells us about their electron arrangements.	Students learned that the periodic law states that the properties of elements repeat in a regular pattern when arranged by increasing atomic number. Elements in the same vertical group share the same number of valence electrons and therefore similar chemical properties. Periods are horizontal rows; groups are vertical columns. The number of electron shells equals the period number; the number of valence electrons equals the group number (for Groups 1–8). Pedagogical note: Emphasise that nitrogen (Group 15, 5 valence electrons) and phosphorus (Group 15, 5 valence electrons) behave similarly — this is why both appear in fertiliser compounds.	Students explain that the repeating pattern of valence electrons is the reason elements in the same group form similar compounds and have similar combining powers. They connect this to the fertiliser phenomenon by reasoning that nitrogen and phosphorus both form compounds with hydrogen and oxygen in similar ways because they share the same number of valence electrons, which determines their reactivity. Teacher notes: Ask 'Why might a Kenyan farmer find that both nitrogen-based and phosphorus-based fertilisers can supply nutrients — but in very different amounts?' Use this to foreshadow valency in Lesson 5.
Lesson 3 Stability, Electron Affinity, and the Noble Gas Rule	Students observe electron shell diagrams of noble gases (He, Ne, Ar) and compare them with sodium (Na) and chlorine (Cl). They note that noble gases have full outer shells (2 or 8 electrons) and are chemically inert, while Na has 1 valence electron and Cl has 7. Teacher notes: WAIT TIME of 10 seconds after 'Why do you think noble gases do not react?' before cold-calling 3 students. Anchor to phenomenon: ask 'If atoms 'want' full outer shells, what might sodium and chlorine do to achieve this — and how could this relate to the ions found in fertilisers?'	Students learned that atoms are most stable when their outer shell is full, matching a noble gas configuration (the octet rule, or duet for the first shell). Metals with few valence electrons tend to lose electrons (low ionisation energy) to achieve stability, while non-metals with many valence electrons tend to gain electrons (high electron affinity). This drive toward stability is the chemical basis for ion formation, bond formation, and ultimately the specific combining ratios seen in fertiliser formulae. Pedagogical note: Reinforce with the Kenyan context — potassium (K, 1 valence electron) in NPK fertiliser loses 1 electron to become K^+, matching argon's configuration.	Students explain that the tendency of atoms to gain or lose electrons in order to achieve a noble gas configuration determines which elements bond together and in what ratios — directly explaining why fertiliser formulae are specific (e.g., K_2SO_4 rather than K_3SO_4). Teacher notes: Prompt with 'How does the noble gas rule explain why the fertiliser formula is fixed and cannot be arbitrarily changed?' Cold-call 2 students and ask them to connect the noble gas rule to one specific element on the fertiliser bag displayed in class.
Lesson 4 Ion Formation: Dot-and-Cross Diagrams	Students observe dot-and-cross diagrams showing electron transfer between sodium and chlorine (forming Na^+ and Cl^-) and between magnesium and oxygen (forming Mg^{2+} and O^{2-}). They note that the number of electrons transferred equals the charge on the ion. Teacher notes: Provide pre-drawn blank dot-and-cross templates. WAIT TIME of 15 seconds after 'What pattern do	Students learned that metallic elements in Groups 1, 2, and 3 lose electrons equal to their group number to form positively charged cations (e.g., Na^+, Mg^{2+}, Al^{3+}), while non-metallic elements in Groups 6 and 7 gain electrons to form negatively charged anions (e.g., O^{2-}, Cl^-). The dot-and-cross diagram is a tool to represent this electron transfer visually, showing the full	Students explain that the specific charges on ions are not arbitrary — they are determined by how many electrons must be transferred to reach a noble gas configuration. This directly explains why fertiliser formulae are fixed: e.g., calcium phosphate is $Ca_3(PO_4)_2$ because Ca forms Ca^{2+} (loses 2 electrons) and PO_4 carries a 3– charge, and the formula is dictated by

	you see between the group number and the charge on the ion?' before cold-calling 4 students. Connect to fertiliser: show that ammonium nitrate (NH_4NO_3) contains the NH_4^+ ion — ask students to predict its charge origin.	outer shells achieved. Pedagogical note: Emphasise that the charge on the ion (1+, 2+, etc.) is precisely the valency used to write fertiliser formulae — making this lesson a direct building block for Lesson 5.	charge balance. Teacher notes: Ask 'Using what you know about ion charges, why can't a farmer simply double the amount of one element in a fertiliser formula?' Cold-call 3 students.
Lesson 5 Valency, Oxidation Numbers, and Radicals: Unlocking the Hidden Pattern in Fertiliser Formulae	Students examine a reference table listing common elements and radicals with their valencies (e.g., N = 3 or 5, P = 3 or 5, K = 1, $\text{NH}_4^+ = 1$, $\text{NO}_3^- = 1$, $\text{SO}_4^{2-} = 2$, $\text{PO}_4^{3-} = 3$). They also observe the formulae on three Kenyan fertiliser bags (CAN, DAP, NPK) and note which radicals appear. Teacher notes: WAIT TIME of 10 seconds after 'What do you notice about the numbers in the formulae compared to the valency table?' before cold-calling 3 students. This is a key sense-making lesson — anchor explicitly back to the driving question.	Students learned that valency is the combining power of an element or radical, determined by electrons lost, gained, or shared. Oxidation numbers follow rules: hydrogen is +1, oxygen is -2, and the sum of oxidation numbers in a neutral compound is zero. Radicals (polyatomic ions such as NH_4^+ , NO_3^- , SO_4^{2-} , PO_4^{3-}) act as single units in formulae. The valency crossover method allows students to write correct formulae: write symbols, place valencies, then cross them over as subscripts (simplifying where needed). Pedagogical note: This is the conceptual core of the sub-strand — invest time ensuring every student can identify a radical and state its valency before moving to formula writing in Lesson 6.	Students explain that the specific and different chemical formulae on Kenyan fertiliser bags (e.g., $(\text{NH}_4)_2\text{SO}_4$ for ammonium sulphate, $\text{Ca}(\text{H}_2\text{PO}_4)_2$ for single superphosphate) are a direct consequence of valency and charge balance — atoms and radicals combine in exact ratios so that total positive valency equals total negative valency. The fertiliser formula cannot be changed without making a compound that does not exist chemically. Teacher notes: Directly address the driving question: 'We can now begin to answer why fertiliser formulae are so specific — ask one student to explain this in one sentence to the class.' Cold-call 2 students.
Lesson 6 Writing Chemical Formulae of Common Compounds	Students observe worked examples of the crossover-valency method applied to binary compounds (e.g., magnesium oxide: Mg^{2+} and $\text{O}^{2-} \rightarrow \text{MgO}$) and compounds containing radicals (e.g., calcium nitrate: Ca^{2+} and $\text{NO}_3^- \rightarrow \text{Ca}(\text{NO}_3)_2$). They then practise writing formulae for at least five compounds independently, including one fertiliser compound. Teacher notes: Circulate and check that students are simplifying ratios correctly (e.g., $\text{Ca}^{2+} + \text{O}^{2-} \rightarrow \text{CaO}$, not Ca_2O_2). WAIT TIME of 10 seconds after each worked example before cold-calling 2 students to verify each step.	Students learned to apply the crossover-valency method systematically: (i) write the symbols of the combining elements or radicals; (ii) write their valencies above; (iii) crossover the valencies as subscripts; (iv) simplify to the lowest whole-number ratio; (v) place brackets around radicals when the subscript is greater than 1. They can write and verify formulae for binary compounds (e.g., Al_2O_3 , MgCl_2) and compounds with radicals (e.g., $(\text{NH}_4)_2\text{SO}_4$, K_2CO_3 , $\text{Ca}_3(\text{PO}_4)_2$). Pedagogical note: Verification using oxidation numbers is a powerful checking tool — teach students to confirm their formula by ensuring the sum of oxidation numbers equals zero.	Students explain that every formula on a fertiliser bag is the only chemically correct formula for that compound, derived from the fixed valencies of its components. For example, diammonium phosphate (DAP) is $(\text{NH}_4)_2\text{HPO}_4$ because NH_4^+ has valency 1 and HPO_4^{2-} has valency 2, requiring two ammonium units to balance one HPO_4 unit. Teacher notes: Ask 'If a fertiliser company tried to make a compound with the formula $(\text{NH}_4)_3\text{SO}_4$, would it be chemically valid? Explain using valency.' WAIT TIME 15 seconds, then cold-call 3 students.
Lesson 7 Balancing Chemical Equations: Writing the Language of Fertiliser Formulae	Students observe an unbalanced equation for the industrial synthesis of ammonia ($\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$ — the Haber process, linking to Kenyan fertiliser imports) and count atoms on each side. They note the	Students learned that: (i) a balanced chemical equation has the same number of each type of atom on both sides, in accordance with the law of conservation of mass; (ii) only coefficients (numbers placed	Students explain that balanced equations represent real chemical events in fertiliser production — the exact ratios of reactants and products are fixed by atomic structure and valency, which is why manufacturers

	imbalance. A second unbalanced equation for the reaction of ammonia with phosphoric acid to form DAP is also examined. Teacher notes: WAIT TIME of 10 seconds after 'What rule is being broken in this unbalanced equation?' before cold-calling 3 students. Emphasise the law of conservation of mass — atoms are neither created nor destroyed.	before formulae) may be changed to balance an equation — subscripts within a formula are fixed by valency and may never be altered; (iii) a systematic approach (balance one element at a time, leave oxygen and hydrogen for last) makes balancing manageable; (iv) state symbols (s), (l), (g), (aq) give additional information. Pedagogical note: The critical misconception to address is students changing subscripts to balance — reinforce firmly that changing a subscript changes the identity of the compound.	follow precise formulae. For example, $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ shows that three molecules of hydrogen are always needed per molecule of nitrogen, a ratio determined by the valency of nitrogen (3) and hydrogen (1). Teacher notes: Anchor to the driving question: 'How does balancing an equation connect the inner structure of an atom — specifically valency — to the specific formula printed on a fertiliser bag in Kericho or Nakuru?' Cold-call 2 students for a final summary response.
Lesson 8 The Periodic Table and Fertiliser Formulae: Pulling It All Together	Students examine a completed anchor chart (or DQB) accumulated across all 8 lessons, displaying the original fertiliser bags alongside student-generated models connecting atomic structure → electron configuration → periodic law → ion formation → valency → formula writing → balanced equations. They observe how each lesson's concept built upon the previous. Teacher notes: Give students 5 minutes of silent review of the anchor chart before discussion. WAIT TIME of 15 seconds after 'Looking at everything we have built, which single concept do you think is most important for explaining the fertiliser formulae — and why?' before cold-calling 4 students.	Students learned that the specific and different chemical formulae on Kenyan fertiliser bags are not arbitrary — they are the direct consequence of a connected chain of atomic-level reasoning: the number of protons defines the element (atomic structure); electrons arrange in shells determining valence electrons (electron configuration); the periodic table organises elements by this repeating pattern (periodic law); atoms gain or lose electrons to achieve stability (noble gas rule and ion formation); the number of electrons transferred defines valency; valency dictates the only valid combining ratio (formula writing); and balanced equations confirm that no atoms are created or destroyed in the process. Pedagogical note: Use this lesson to formally revisit and answer the driving question as a whole class. Every student should be able to give a complete, multi-step explanation.	Students explain the full answer to the driving question: fertiliser bags carry specific and different chemical formulae because each element's atomic structure — particularly the number and arrangement of its electrons — determines how many electrons it can lose, gain, or share (its valency), which in turn dictates the exact ratio in which atoms must combine to form a stable, electrically neutral compound. Changing the formula would violate the laws of charge balance and conservation of mass. Teacher notes: As a culminating task, ask each student to write one paragraph answering the driving question using at least five key vocabulary terms (atomic number, electron configuration, valency, ion, balanced equation). Cold-call 3 students to read their paragraph aloud before collecting written responses as a formative assessment.